

1. (Exercise 2.21) A fair coin is flipped 3 times. The toss results are recorded on separate slips of paper (writing "H" if Heads and "T" if tails), and the 3 slips of paper are thrown into a hat.

(a) Find the probability that all 3 tosses landed Heads, given that at least 2 were Heads.

$P(A)$ = all 3 tosses are heads

$P(B)$ = at least 2 tosses are heads

$P(A|B) = \frac{P(A \cap B)}{P(B)}$ → Prob. that all 3 landed H given that at least 2 were H early.

$$P(A) = \frac{\text{Want:}}{\text{Choices}} = \frac{(H) 1}{(H,T) 2} \cdot \frac{(H) 1}{(H,T) 2} \cdot \frac{(H) 1}{(H,T) 2} = \frac{1 \cdot 1 \cdot 1}{2 \cdot 2 \cdot 2} = \frac{1}{8}$$

$$P(A) = \frac{1}{8} \rightarrow \text{all 3 HEADS}$$

$$P(B) = \left\{ \begin{array}{l} H, H, H \rightarrow \frac{1}{8} \\ H, H, T \rightarrow \frac{1}{8} \\ H, T, H \rightarrow \frac{1}{8} \\ T, H, H \rightarrow \frac{1}{8} \end{array} \right.$$

$$\Rightarrow P(B) = 4 \cdot \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P(A \cap B) = \{H, H, H\} \cap \left\{ \begin{array}{l} H, H, H \\ H, H, T \\ H, T, H \\ T, H, H \end{array} \right\} = \left[\frac{1}{8} \right]$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{8}}{\frac{1}{4}} = \frac{4}{8} = \frac{1}{2}$$

$$P(A|B) = \frac{1}{4} = 25\%$$

(b) Two of the slips of paper are randomly drawn from the hat, and both show the letter H. Given this information, what is the probability that all 3 tosses landed Heads?

A $\rightarrow$ all 3 tosses are Heads

B $\rightarrow$ 2 slips from hat are Heads

$P(A|B) = \frac{\text{Prob. that all 3 are Heads given that 2 slips drawn are 2 (Now)}}{1}$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} \rightarrow \begin{array}{l} \text{2 slips are Head (now)} \\ \text{could be also 3 heads later.} \end{array}$$

$P(B|A)$ = Probability getting 2 Heads slips given than there are 3 Heads tails
= Probability $\rightarrow 1$

$$P(B|A) = 1$$

$P(A)$ = all 3 are Heads = $\frac{1(H)}{2(H,T)} \cdot \frac{1(H)}{2(H,T)} \cdot \frac{1(H)}{2(H,T)} = \frac{1}{8}$

$$P(A) = \frac{1}{8}$$

$P(B)$ $\rightarrow$ can be 2 events:

1.) 2 heads and 1 tail $\Rightarrow$ $\begin{cases} HHT \\ HTH \\ T HH \end{cases}$ $\rightarrow \frac{1}{8}, \frac{1}{8}, \frac{1}{8}$
 $3 \cdot \frac{1}{8} = \frac{3}{8}$

1.) $\frac{3}{8}$

2.) all 3 Heads

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

2 2 2 8

2.) $\frac{1}{8}$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)} =$$

$$= \frac{1 \cdot \frac{1}{8}}{\frac{1}{8} + \frac{3}{8}} = \frac{\frac{1}{8}}{\frac{4}{8}} = \frac{1}{4} = \frac{8}{32} = \frac{1}{4}$$

$$P(A|B) = \frac{1}{4} = 25\%$$

2. (Exercise 2.8) The screens used for a certain type of cell phone are manufactured by 3 companies, A, B, and C. The proportions of screens supplied by A, B, and C are 0.5, 0.3, and 0.2, respectively, and their screens are defective with probabilities 0.01, 0.02, and 0.03, respectively. Given that the screen on such a phone is defective, what is the probability that Company A manufactured it?

Supplied given Company

$$P(S|A) = 0.5$$

$$P(S|B) = 0.3$$

$$P(S|C) = 0.2$$

defective given Company

$$P(D|A) = 0.01$$

$$P(D|B) = 0.02$$

$$P(D|C) = 0.03$$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$P(A|D) = ?$ Prob. that its made by compny A,
Given that its Defective.

$$P(A|D) = \frac{P(D|A) \cdot P(A)}{P(D)} \rightarrow \begin{matrix} P(\text{supplied by A}) \\ P(\text{Defective all}) \end{matrix}$$

$$P(A|D) = \frac{(0.01) \cdot 0.5}{0.01 \cdot 0.5 + 0.02 \cdot 0.5 + 0.03 \cdot 0.2} =$$

$$= \frac{\frac{1}{200}}{\frac{17}{1000}} = \frac{\overset{5}{1000}}{200 \cdot 17} = \frac{5}{17} = 0.294 = 29\%$$

$$P(A|D) = 29\%$$